Solutions

Worksheet 1: August 28th

Problem 1. Let n and m be integers. We say that n *divides* m, written $n \mid m$, if there exists some other integer k such that $m = kn$.
Let a, b, c be integers. Suppose $a \mid b$ and $a \mid c$. Prove that for every other two integers m, n,

$$a \mid (mb + nc).$$

By assumption, $b = ka$ and $c = la$ for integers k, l.
Then $mb + nc = mka + nla$
$= (mk + nl)a$ for $(mk+nl) \in \mathbb{Z}$,

and so $a \mid mb+nc$.

Problem 2. Prove that if n^2 is even, then n is even.

If n is odd, $n = (2k+1)$ for $k \in \mathbb{Z}$. Then,
$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is also odd.
By contraposition, the claim is true.

Problem 3. We say that a positive integer n is prime if the only positive integers which divide it are 1 and n. Prove by induction that every positive integer ≥ 2 can be written as the product of primes.

We apply induction. First, notice 2 is prime, and is trivially such a product. Assume that every number $2 \leq k \leq n$ is a product of primes. Then, $n+1$ is either prime, in which case it is a product of primes, or $n+1 = a \cdot b$ for $a, b > 1$. Then, $a, b \leq n$ are products of primes, and so $n+1$ is,

Problem 4. Let F_n denote the n-th Fibonacci number which is defined recursively via

$$F_0 = 0, \qquad F_1 = 1, \qquad F_n = F_{n-1} + F_{n-2}.$$

Show by induction that $F_n < 2^n$.

Base Case: $F_1 = 1 \leq 2 = 2^1$.

Inductive Case: Assume $F_k \leq 2^k$ for $k \leq n$. Then,

$F_{n+1} = F_n + F_{n-1} \leq 2^n + 2^{n-1} \leq 2^{n+1}$.

Problem 5. Let $f : X \to Y$ and $g : Y \to Z$ be functions.

(a) Prove that if $g \circ f$ is injective, then f is injective.

If $f(a) = f(b)$, then $g(f(a)) = g(f(b))$ and so $a = b$.

(b) Is the converse true? If it is, prove it. If not, provide a counterexample.

No, if g fails to be injective

$f(x) = x$, $g(x) = 2$.